

# **ARRAYS**

**It is collection of homogeneous [same type] variables.**

**Array is nothing but collection of contiguous memory locations, where we can store and manage more than one value of same type under one name.**

**It is a derived data type.**

**It is an implicit / internal pointer.**

**It is an implicit const pointer**

**It is one of data structure.**

## **Advantages:**

**Generally to store several values of same type, we have to declare several variables. Here we have to remember all these variable names also. When the program is too big, it is very difficult to remember all the variable names. In this situation, the only solution is array.**

**Array reduce program length.**

**Array minimize the errors.**

**In functions to carry several values of same type at a time, we are using arrays.**

**It allows to arrange our data in a order.**

## **Disadvantage:**

Array size is Constant Positive Integer value. Due to this we are not able to change the array size at run time. Sometimes it causes memory wastage / shortage.

In C language we are using

1. One dimensional arrays
2. Multi dimensional arrays

### One dimensional arrays:

- An array with one row and several columns.
- An array with single subscripting operator **[ ]** is called one dimensional array.
- It is an implicit single pointer.

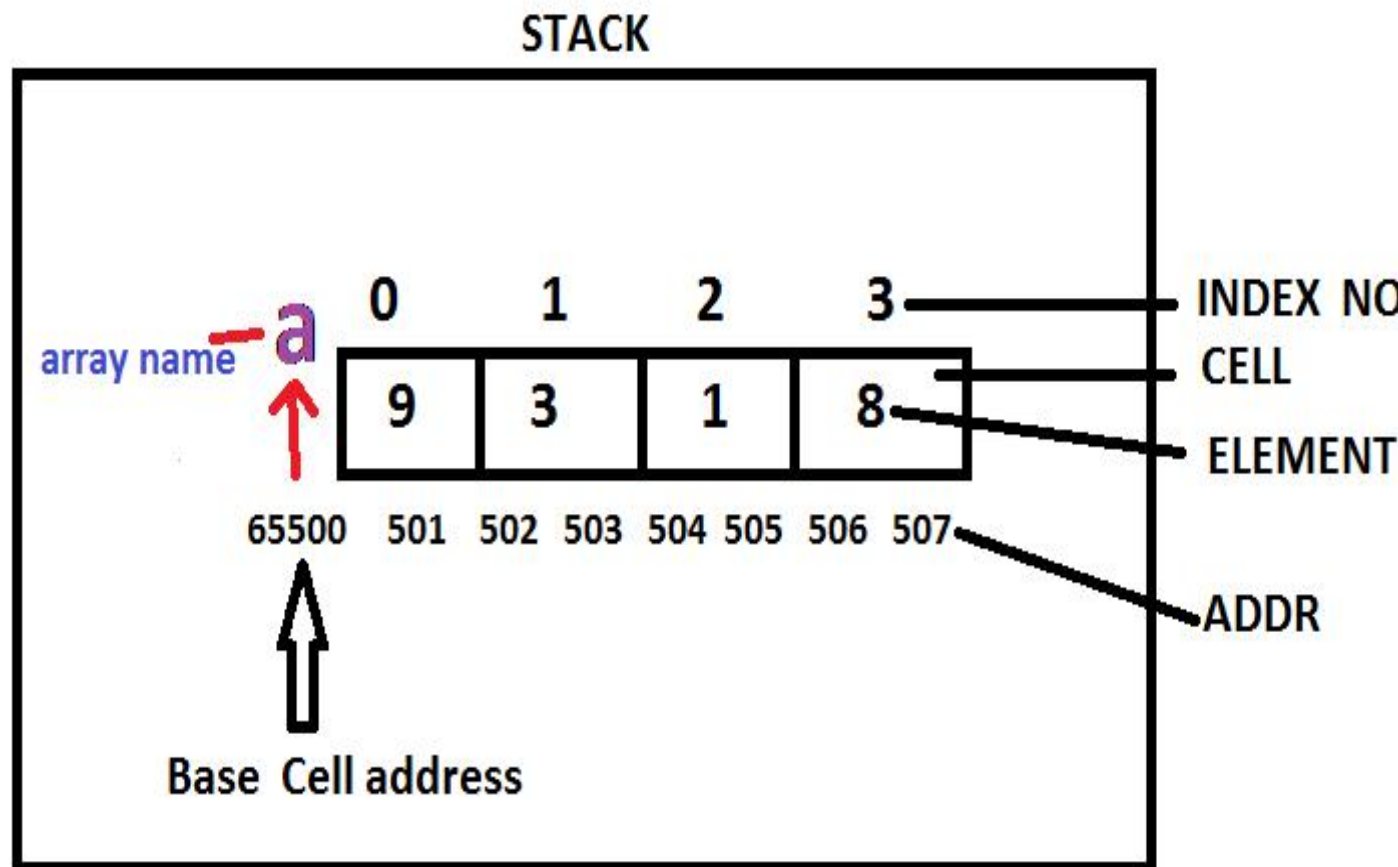
### Syntax:

**datatype variable[size] = {elements};**

**Eg:**

```
int a[4] = { 9, 3, 1, 8 };
```

### Memory allocation for array:



Array is implicit pointer because of array variable stores base cell [ 0 cell 1<sup>st</sup> byte ] address. Hence array variable value and 0 cell address both are same.

### Array declaration methods:

`int a[3];` Ok

`int a[ ];` No

`int a[3]={1,2,3};` Ok

`int a[ ]={1,2,3};` Ok

**int a[0]={1,2,3}; Ok**

**int a[-5]; No**

**int a[5.5]; No**

**int n = 5, a[n]; No**

**int a[3]={10,20}; Ok**

**int a[3]={1, 2, 3, 4}; No**

**int a[0]; error**

**#define n 5 /\* macro \*/**

**int a[n]; Ok**

**const int n=5, a[n]; No**

**int a[5>3]; → int a[1]; Ok**

**int a[3<2]; → int a[0]; No**

**int a[2+3]; → int a[5]; Ok**

**int a[5%3]; → int a[2]; Ok**

**int a[5%5]; → int a[0]; No**

**int a[1,2,3]; → error**

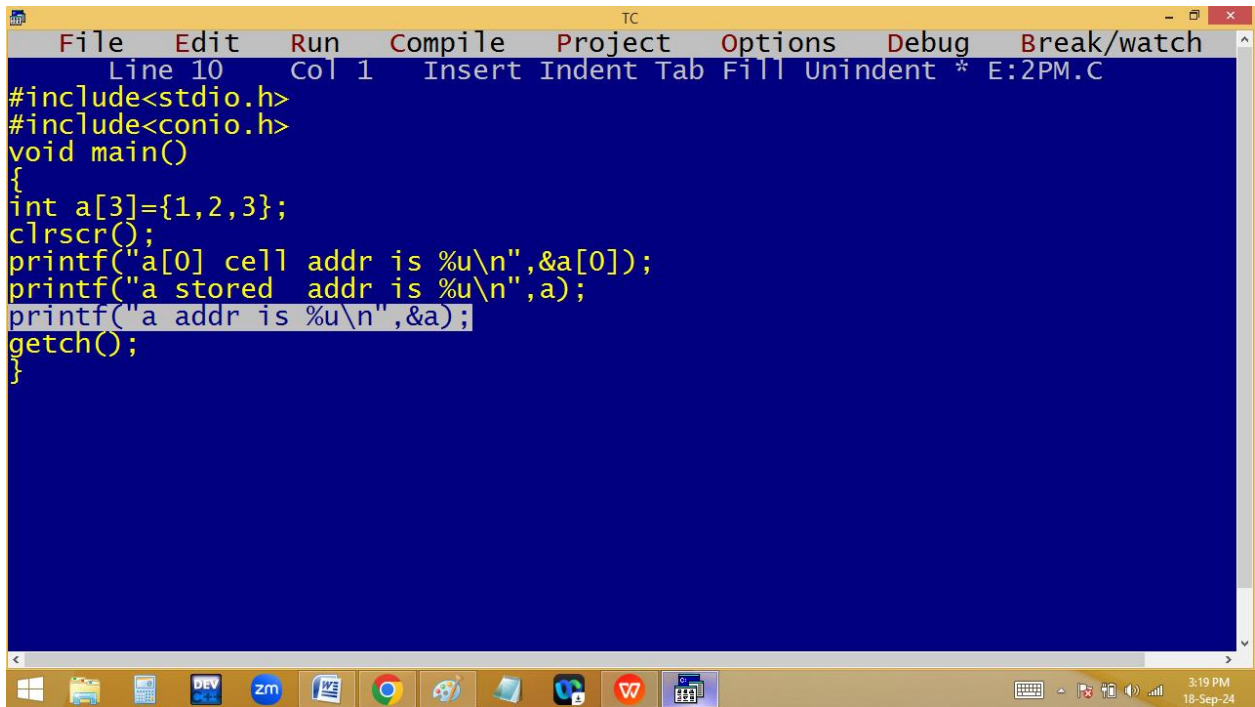
**int a[40000]; → 40000 \* 2 = 80000 bytes → No**

**Note:** Stack size is 65536 bytes(64kb) Only.

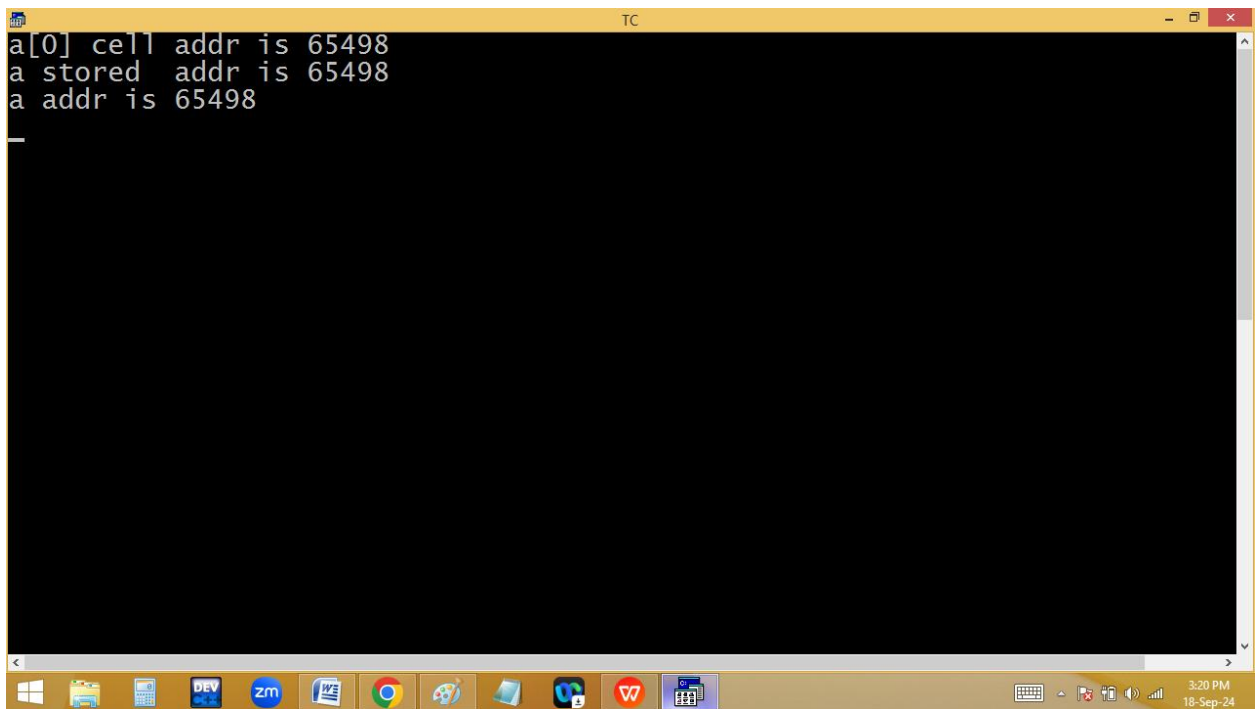
**float a[10000]; Ok → 10000 \* 4 = 40000 bytes**

**float a[20000]; No → 20000 \* 4 = 80000 bytes**

## Finding address of array:

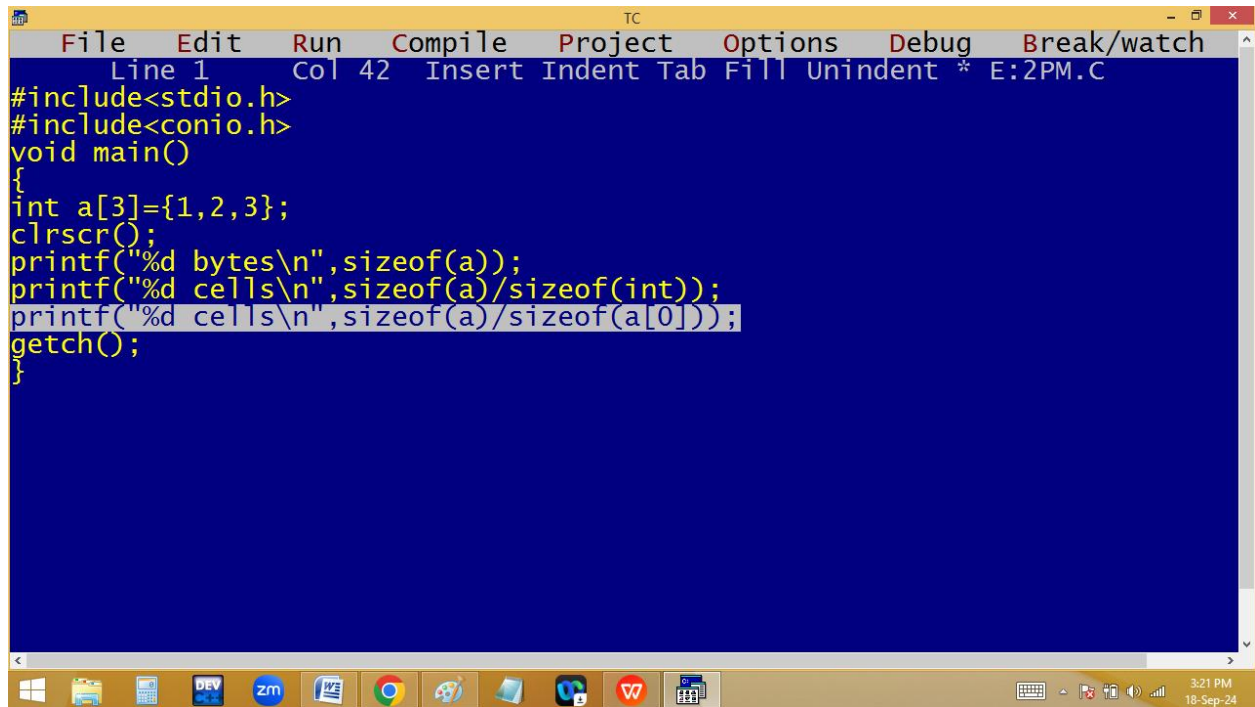


```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[3]={1,2,3};
clrscr();
printf("a[0] cell addr is %u\n",&a[0]);
printf("a stored addr is %u\n",a);
printf("a addr is %u\n",&a);
getch();
}
```

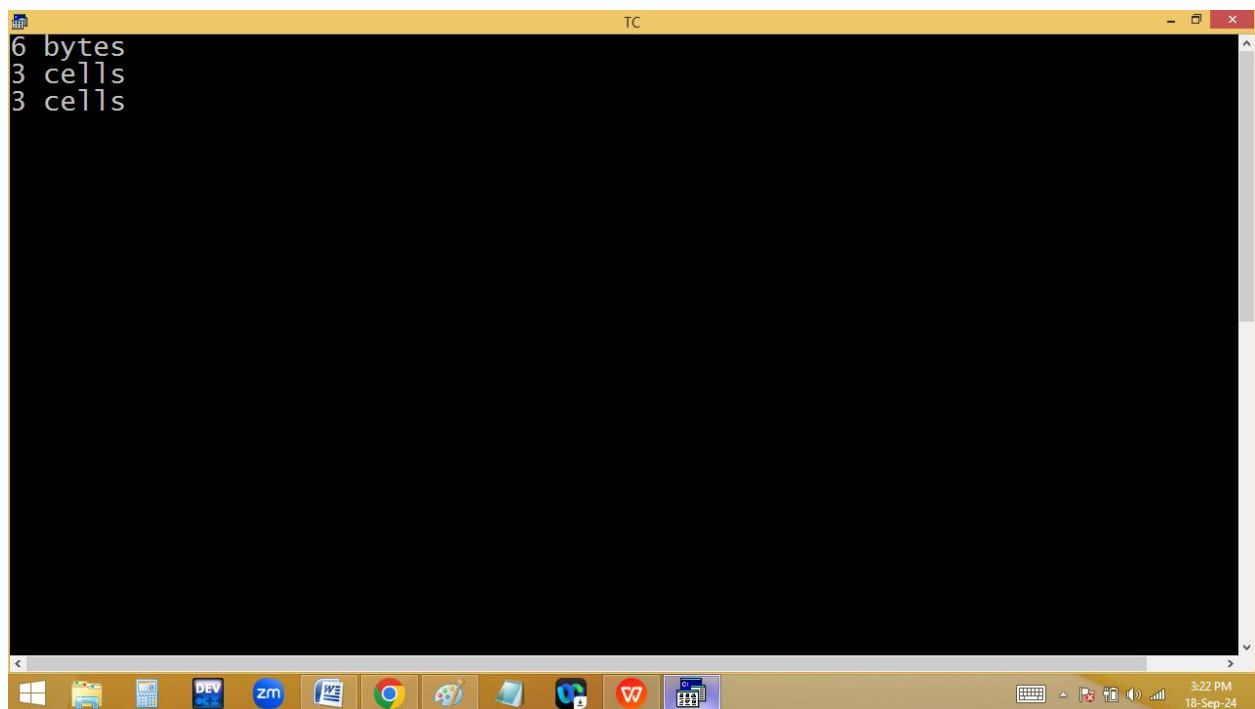


```
TC
a[0] cell addr is 65498
a stored addr is 65498
a addr is 65498
```

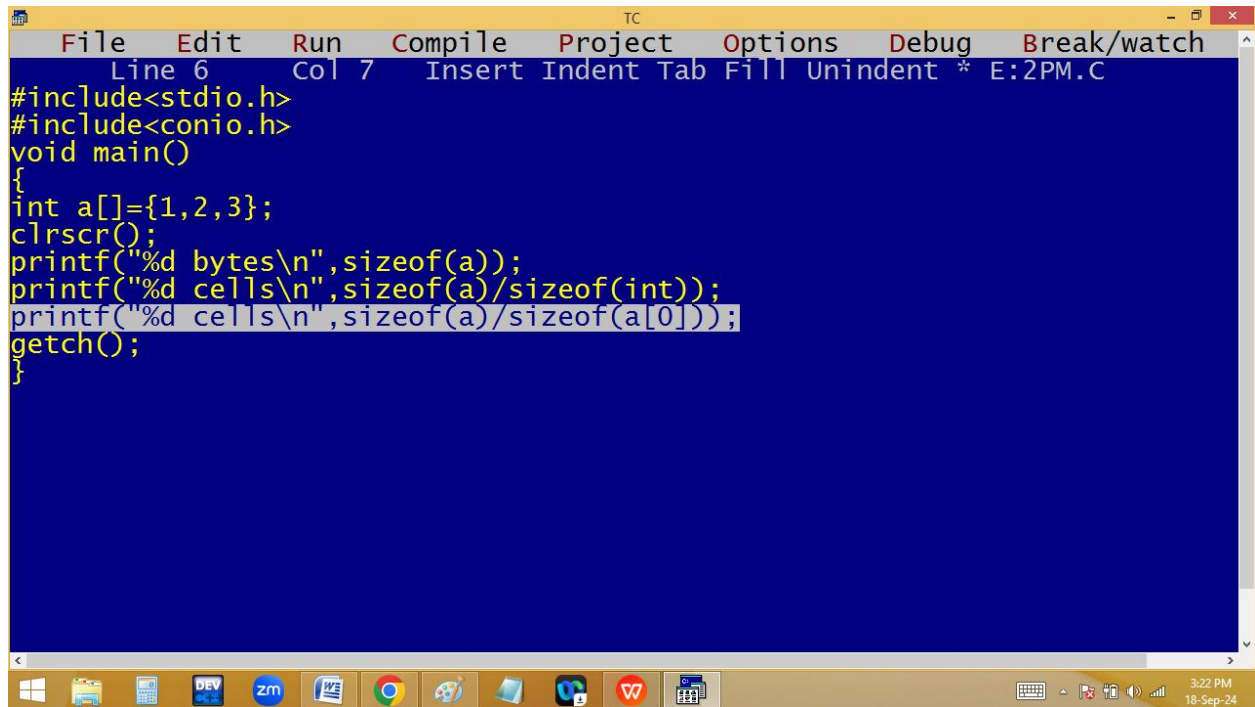
## Finding array size:



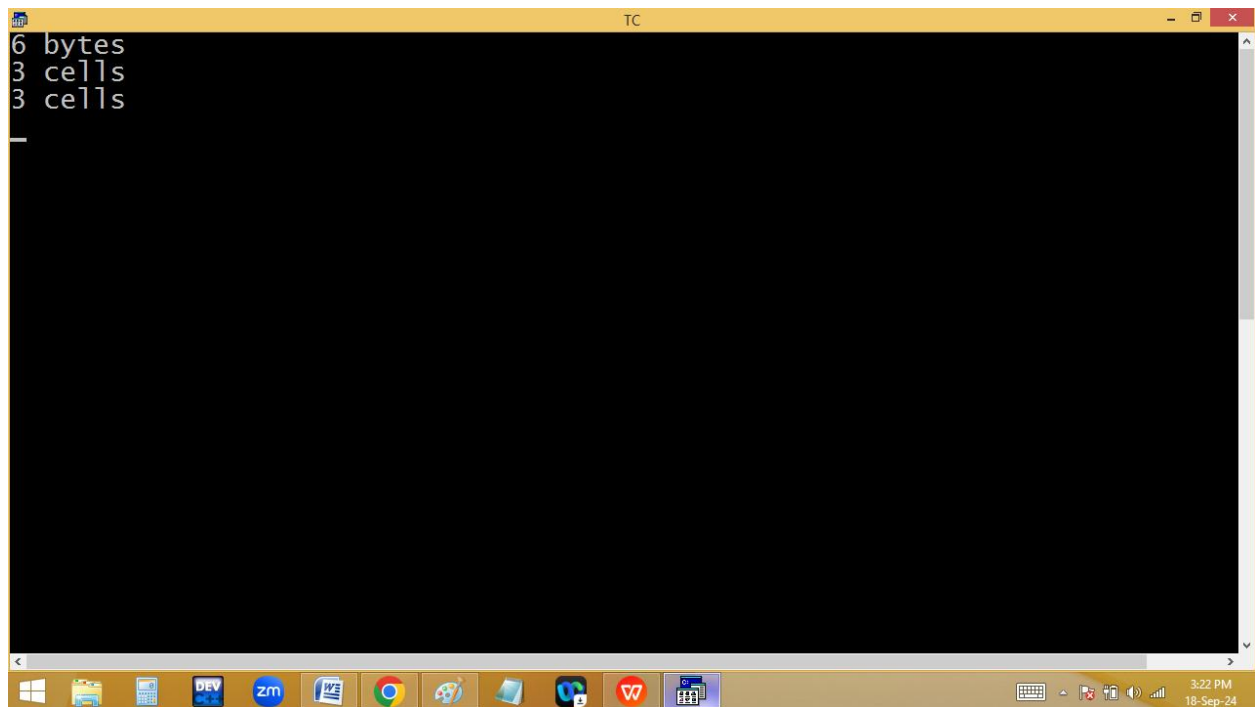
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 1 Col 42 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[3]={1,2,3};
clrscr();
printf("%d bytes\n",sizeof(a));
printf("%d cells\n",sizeof(a)/sizeof(int));
printf("%d cells\n",sizeof(a)/sizeof(a[0]));
getch();
}
```



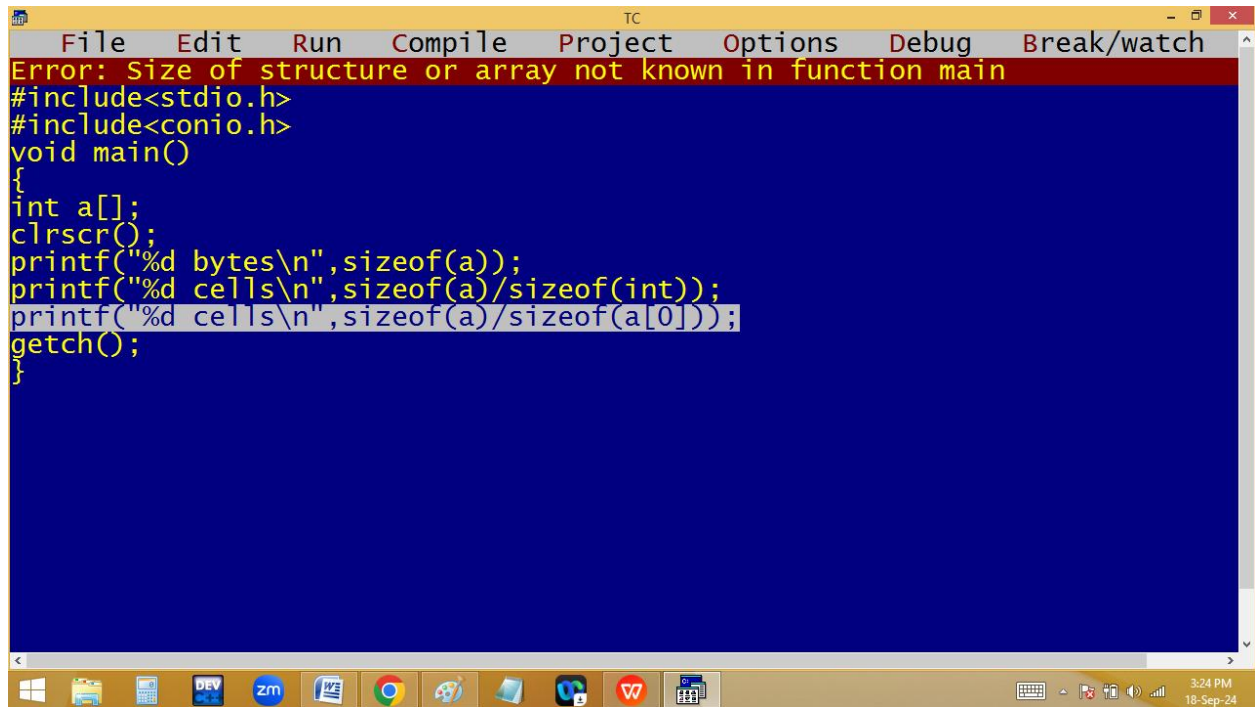
```
TC
6 bytes
3 cells
3 cells
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 6 Col 7 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[]={1,2,3};
clrscr();
printf("%d bytes\n",sizeof(a));
printf("%d cells\n",sizeof(a)/sizeof(int));
printf("%d cells\n",sizeof(a)/sizeof(a[0]));
getch();
}
```



```
TC
6 bytes
3 cells
3 cells
```



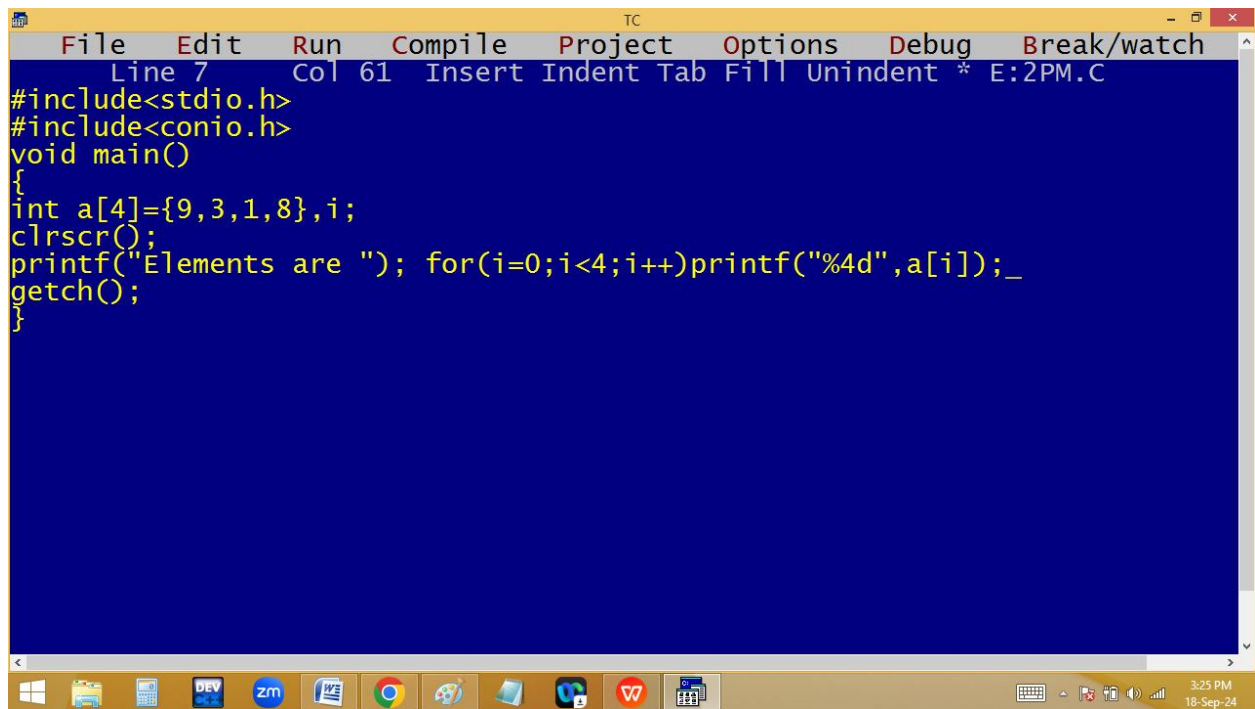
The screenshot shows a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". A red error message banner at the top states: "Error: Size of structure or array not known in function main". The code editor contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[];
clrscr();
printf("%d bytes\n",sizeof(a));
printf("%d cells\n",sizeof(a)/sizeof(int));
printf("%d cells\n",sizeof(a)/sizeof(a[0]));
getch();
}
```

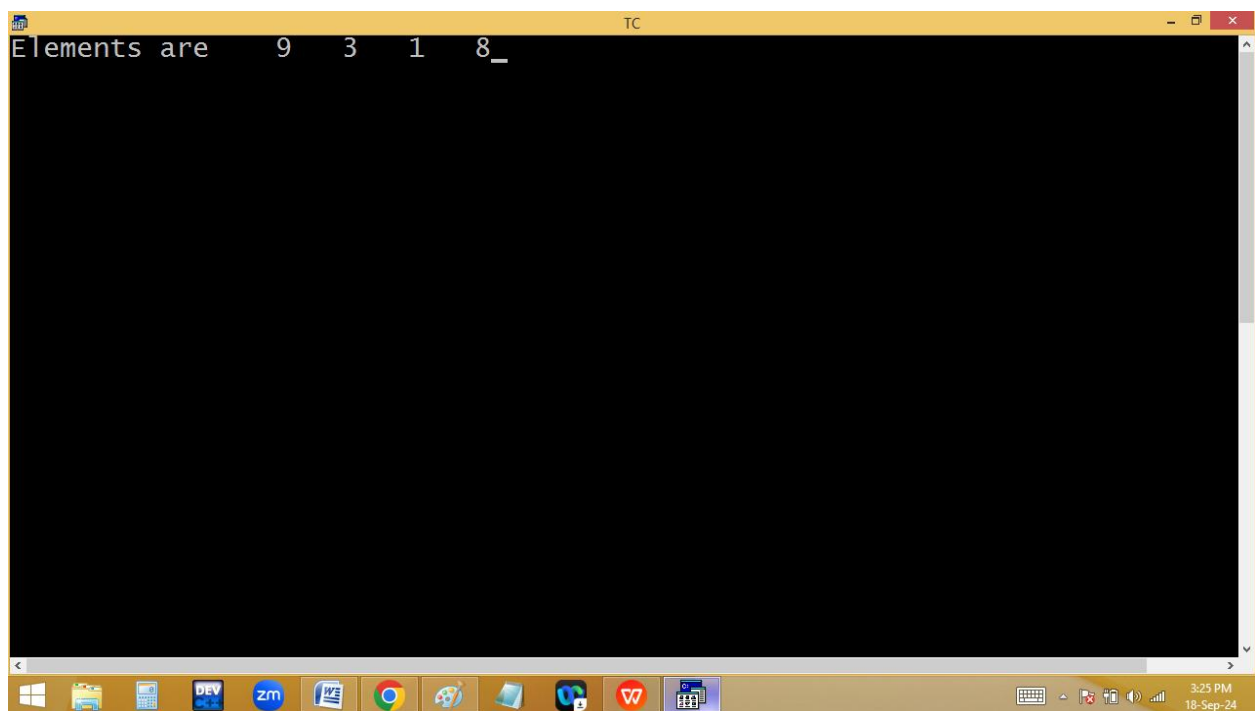
The taskbar at the bottom shows various application icons including Windows Explorer, DEV C++, Zoom, Word, Chrome, and others. The system clock in the bottom right corner indicates 3:24 PM on 18-Sep-24.

**Eg: Direct initialization of array elements:**





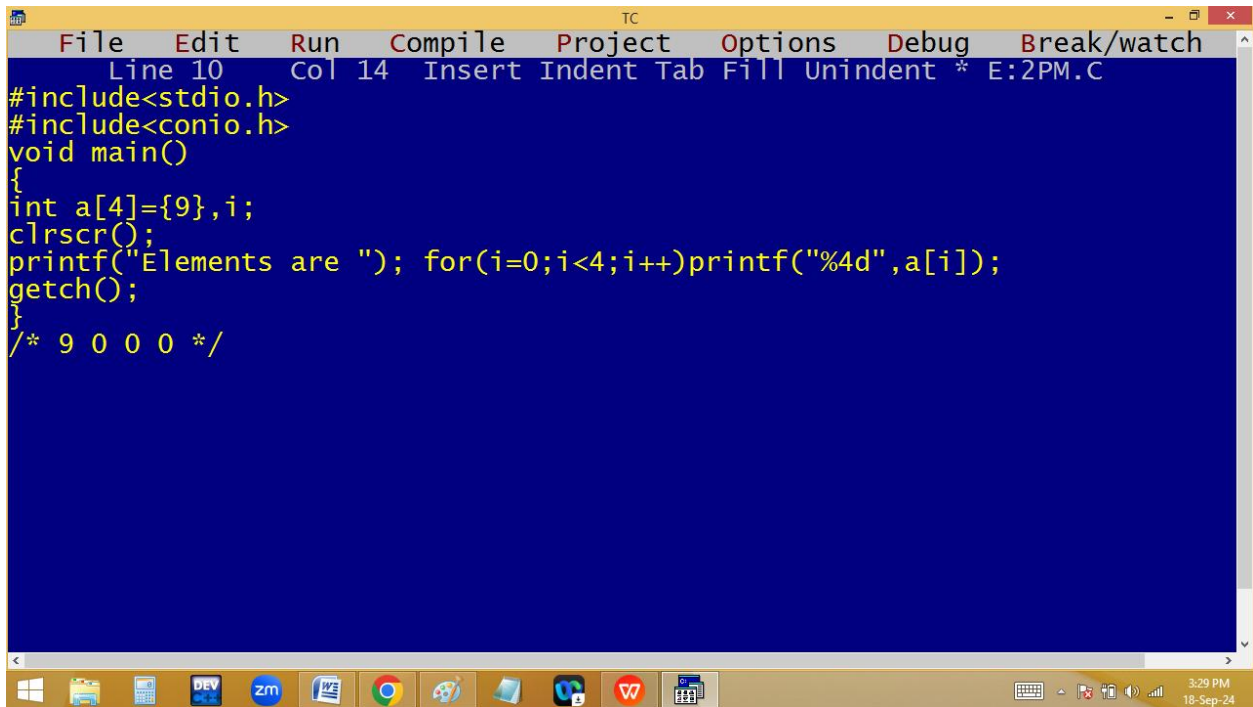
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 61 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[4]={9,3,1,8},i;
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%4d",a[i]);_
getch();
}
```



```
TC
Elements are 9 3 1 8_
```

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 61 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[4]={9,3,1,8},i;
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%4d",i[a]);_
getch();
}
```

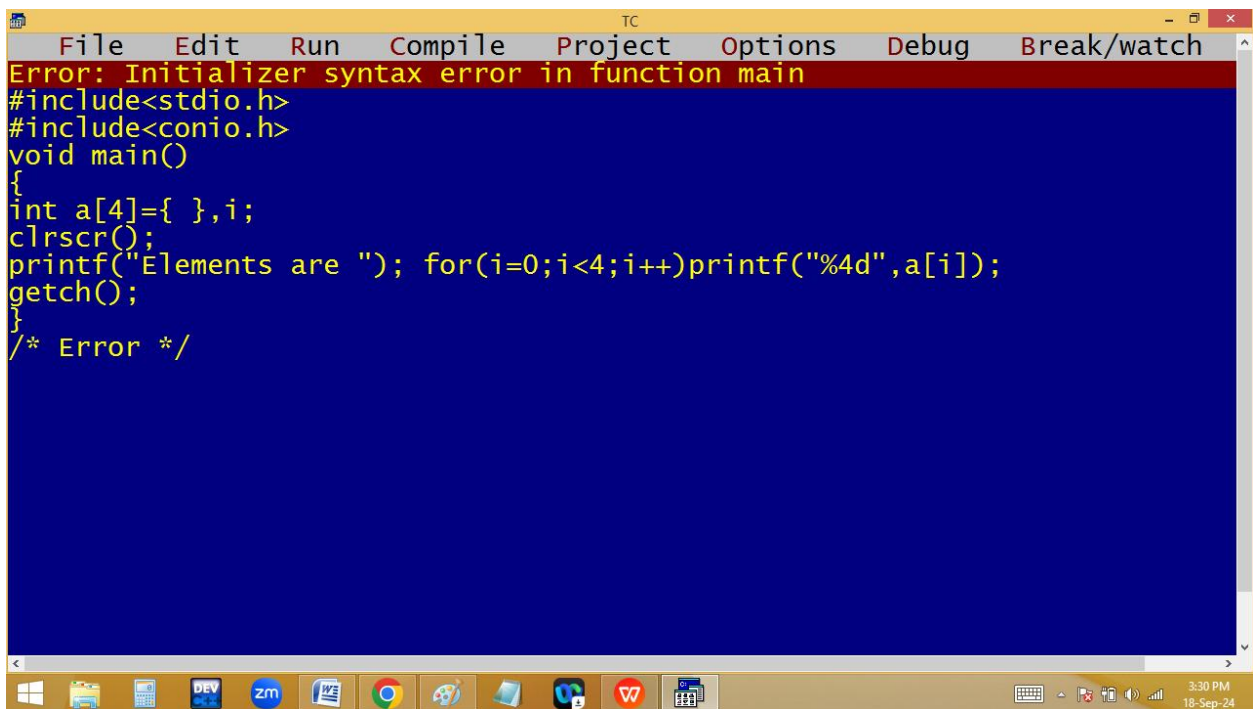
```
TC
Elements are 9 3 1 8_
```



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 10, Col 14, Insert, Indent, Tab, Fill, Unindent, \* E:2PM.C). The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[4]={9},i;
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%4d",a[i]);
getch();
}
/* 9 0 0 0 */
```

The Windows taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, WPS, Chrome, and others. The system clock indicates 3:29 PM on 18-Sep-24.



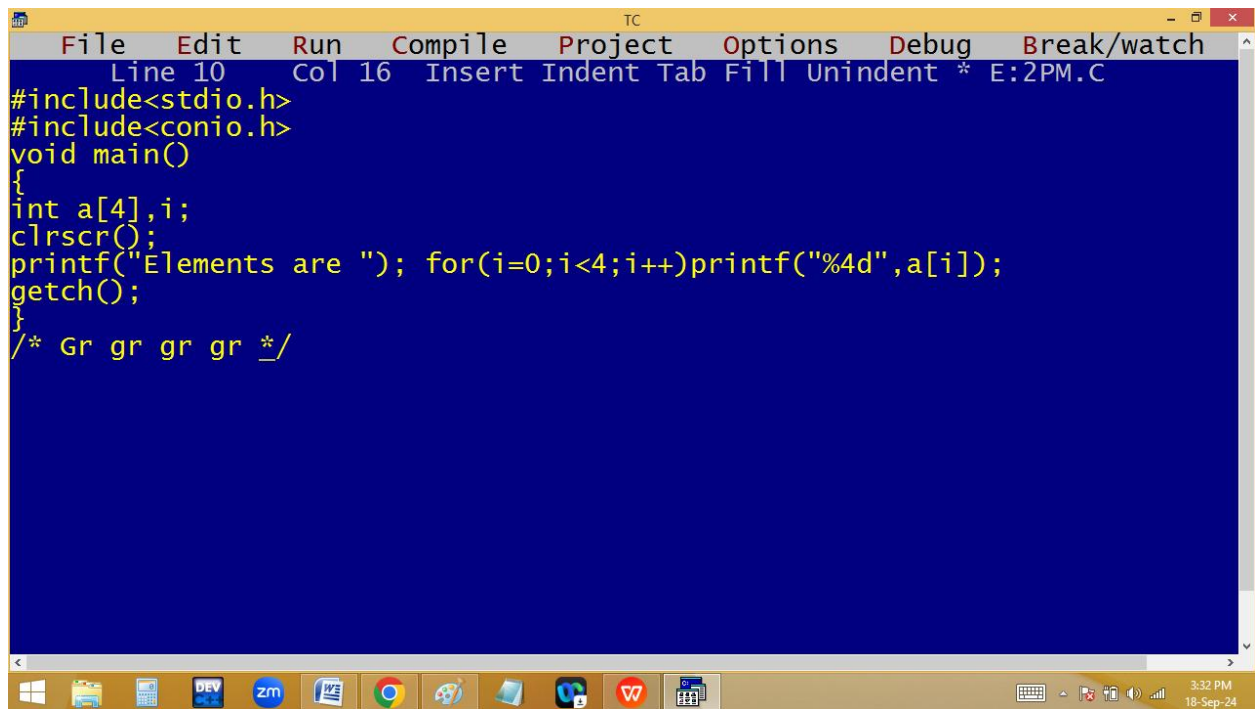
The screenshot shows the same Turbo C++ (TC) IDE window, but with a red error message at the top: "Error: Initializer syntax error in function main". The code in the editor is now:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[4]={ },i;
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%4d",a[i]);
getch();
}
/* Error */
```

The Windows taskbar and system clock remain the same, showing 3:30 PM on 18-Sep-24.

```
TC
File Edit Run Compile Project Options Debug Break/watch
Error: Incompatible type conversion in function main
#include<stdio.h>
#include<conio.h>
void main()
{
int a[4]=9,i;
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%4d",a[i]);
getch();
}
/* Error */
```

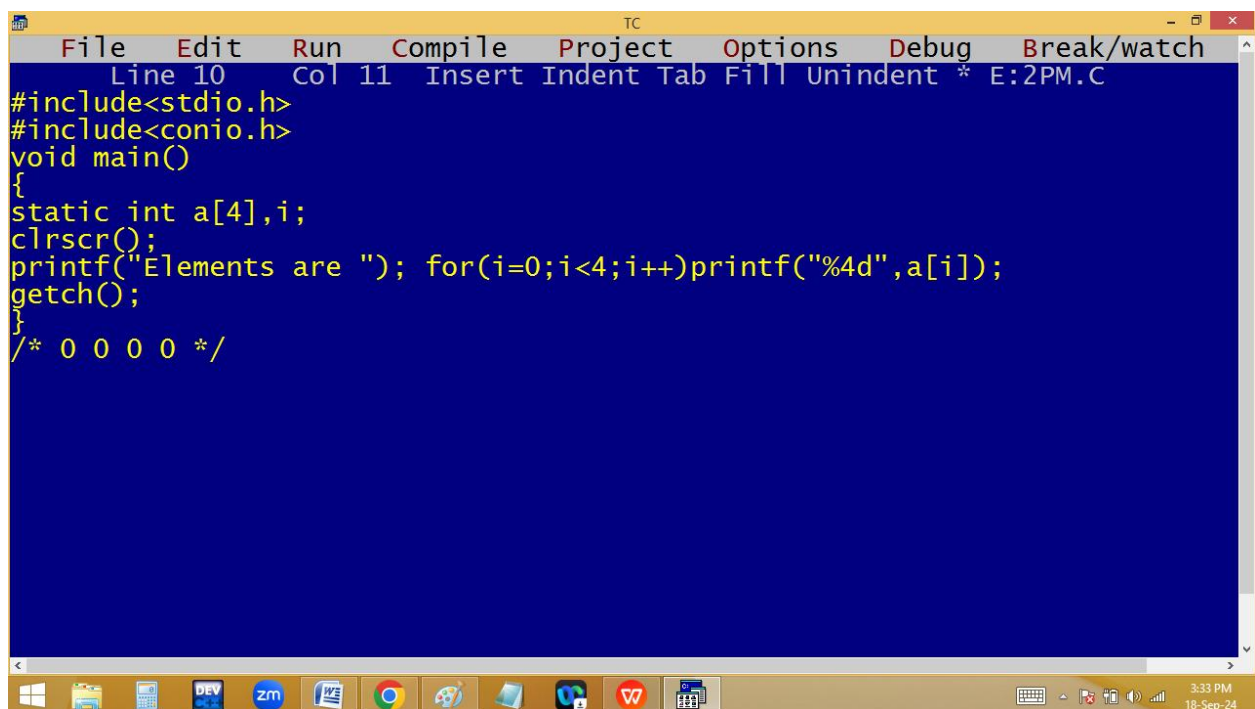
```
TC
File Edit Run Compile Project Options Debug Break/watch
Error: Too many initializers in function main
#include<stdio.h>
#include<conio.h>
void main()
{
int a[4]={9,3,1,8, 4},i;
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%4d",a[i]);
getch();
}
/* Error */
```



The screenshot shows the Turbo C++ (TC) IDE with a blue background. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 10 Col 16 Insert Indent Tab Fill Unindent \* E:2PM.C'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[4],i;
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%4d",a[i]);
getch();
}
/* Gr gr gr gr _/
```

The Windows taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, WPS, Chrome, and others. The system clock in the bottom right corner shows 3:32 PM on 18-Sep-24.



The screenshot shows the Turbo C++ (TC) IDE with a blue background. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 10 Col 11 Insert Indent Tab Fill Unindent \* E:2PM.C'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
static int a[4],i;
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%4d",a[i]);
getch();
}
/* 0 0 0 0 */
```

The Windows taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, WPS, Chrome, and others. The system clock in the bottom right corner shows 3:33 PM on 18-Sep-24.

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 6 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
int a[4],i;
void main()
{
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%4d",a[i]);
getch();
}
/* 0 0 0 0 */
```

```
TC
File Edit Run Compile Project Options Debug Break/watch
Error: Initializer syntax error in function main
#include<stdio.h>
#include<conio.h>

void main()
{
int a[4]={1,2},{3,4}},i;
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%4d",a[i]);
getch();
}
/* Errors */
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 11 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>

void main()
{
int a[4]={1},{2},{3},{4},i;
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%4d",a[i]);
getch();
}
/* 1 2 3 4_*/
```

```
TC
File Edit Run Compile Project Options Debug Break/watch
Error: Expression syntax in function main
#include<stdio.h>
#include<conio.h>

void main()
{
int a[4]={1,2, ,4},i;
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%4d",a[i]);
getch();
}
/* Errors */
```

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 15 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>

void main()
{
int a[4]={ '1', '2', '3', '4'},i;
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%4d",a[i]);
getch();
}
/* 49 50 51 52 */
```

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 11 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>

void main()
{
int a[4]={1.1,2.2,3.3,4.4},i;
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%4d",a[i]);
getch();
}
/* 1 2 3 4_*/
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 16 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>

void main()
{
int a[4]={1>2, 3>1, 5/2, 40000},i;
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%8d",a[i]);
getch();
}
/* 0 1 2 -25536_*/
```

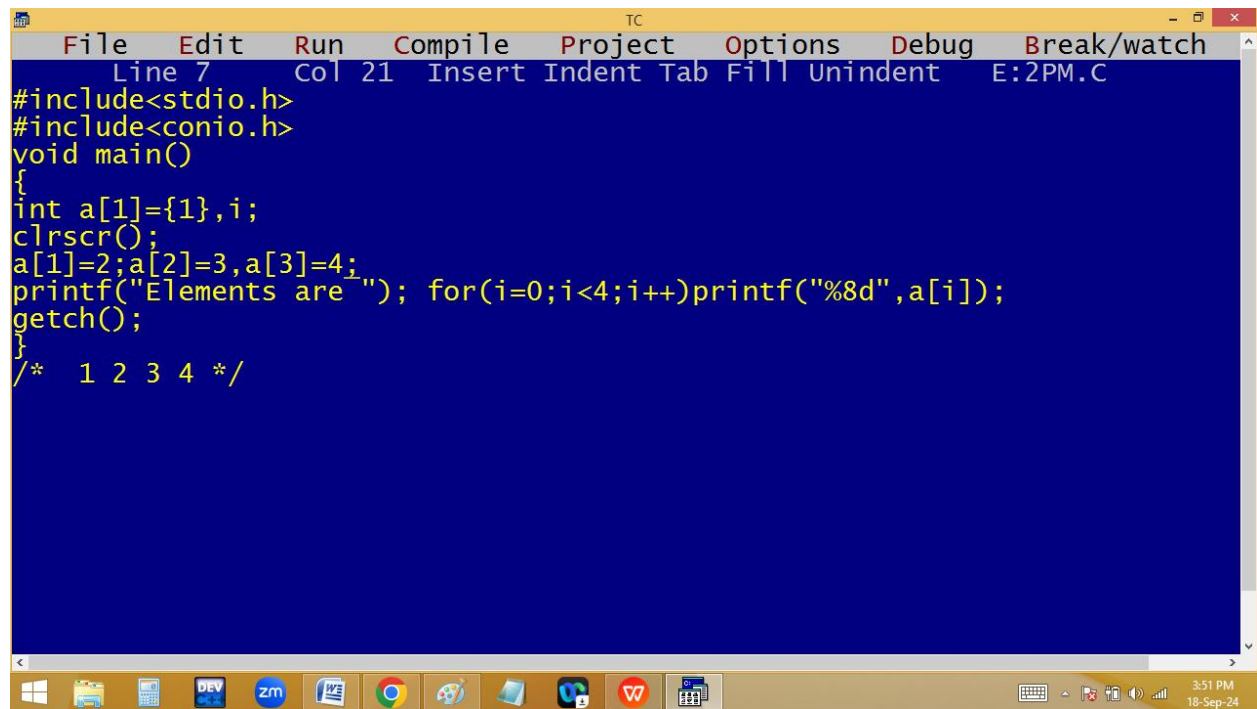
```
TC
File Edit Run Compile Project Options Debug Break/watch
Error: Constant expression required in function main
#include<stdio.h>
#include<conio.h>

void main()
{
int n=4, a[n]={1,2,3,4},i;
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%8d",a[i]);
getch();
}
/* Error */
```

```
TC
File Edit Run Compile Project Options Debug Break/watch
Error: Constant expression required in function main
#include<stdio.h>
#include<conio.h>

void main()
{
const int n=4, a[n]={1,2,3,4},i;
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%8d",a[i]);
getch();
}
/* Error */
```

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 12 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#define n 4 /* macro - constant */
void main()
{
int a[n]={1,2,3,4},i;
clrscr();
printf("Elements are "); for(i=0;i<4;i++)printf("%8d",a[i]);
getch();
}
/* 1 2 3 4_*/
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 21 Insert Indent Tab Fill Unindent E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[1]={1},i;
clrscr();
a[1]=2;a[2]=3,a[3]=4;
printf("Elements are "); for(i=0;i<4;i++)printf("%8d",a[i]);
getch();
}
/* 1 2 3 4 */
```